Melisa YALÇINKAYA

Junior Software Engineer

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SUMMARY

Junior software engineer, graduated in Software Engineering from Atatürk University in 2026. Builds LLM-based assistants (structured output, RAG, agents, personal data masking for KVKK/GDPR) and Turkish BERT models that run in the browser, and end-to-end apps with Django, FastAPI, React and Flutter. Chooses models and prompts by measuring them, and ships projects with automated tests, GitHub Actions CI and Docker.

WORK EXPERIENCE

AI Developer 09/2026 – 10/2026

BRN Teknoloji *Remote, part-time*

- Built an AI assistant for the internal communication of a medical products exporter that turns meeting transcripts into structured records and tracks tasks across meetings (OpenAI API, Pydantic, Streamlit).
- Built the layer that pseudonymises personal data before it reaches OpenAI (zero leaks in evaluation), an audit trail and two-way sync with Notion.
- Added source-cited search over the meeting archive (RAG) and a read-only agent; chose search methods and prompt versions by measuring them.

Intern 07/2025 – 09/2025

Berk Yazılım / Barkodcu *Erzurum, Türkiye*

- At a company building restaurant automation and barcode solutions, worked on the full-stack development of the "QR Menu" application.
- Built web-based customer interfaces, backend API services and desktop management tools that digitalised customer-facing operations.

EDUCATION

BSc, Software Engineering 2022 – 2026

Atatürk University *Erzurum, Türkiye*

- Graduated. Completed the first year at Gümüşhane University (2021 – 2022) and transferred based on GPA for the final three years.

AI Engineering Bootcamp 2026

Tirendaz Akademi *Online, live*

- LLMs, prompt and context engineering, evaluation, embeddings and RAG, AI agents, MCP, fine-tuning and deployment. Capstone: Corporate Communication Assistant.

SKILLS

Programming Languages: Python, JavaScript, Dart, HTML5, CSS

AI: Machine Learning, LLMs and Prompt Engineering, OpenAI API, RAG, Agents (tool calling), PyTorch, Hugging Face Transformers, ONNX, scikit-learn, NLP

Web & Mobile: Django, Django REST Framework, FastAPI, React.js, Flutter, RESTful APIs, JWT, Chrome extensions (MV3)

Tools: Git, GitHub, GitHub Actions (CI), Docker, pytest, Streamlit, Notion API, PySide6, Figma, Unity, Claude

PROJECTS

Corporate Communication Assistant

Technologies: Python, OpenAI API, Pydantic, RAG, Streamlit, Notion API, Docker

- Built a human-in-the-loop LLM app (transcript → personal data masking → LLM draft → staff approval → record + Notion) in which every item cites the transcript lines it is based on.
- Measured prompt versions with five runs on each of three synthetic scenarios; the latest is 100% flawless on the noisy one. Compared embedding, BM25+ and hybrid search for the archive (hit@1 80%); the agent picked the right tool for 14/14 questions. 86 automated tests.

LockNet: Turkish Offensive Text Filter (Chrome extension)

Technologies: PyTorch, Transformers, ONNX Runtime Web, JavaScript, Chrome MV3, GitHub Actions

- Fine-tuned a Turkish BERT on OffensEval-2020 and shipped it inside the extension as int8 ONNX (41 MB): 0.804 macro F1 on the test set (BERTurk 0.821), ~18 ms per sentence; no text leaves the device.
- AUC 0.966 on a hand-labelled set of 478 web texts with 1–2% false alarms on random text. Python/JavaScript preprocessing and tokenizer parity verified by CI tests; found and fixed int8 scores depending on batch neighbours. In Chrome Web Store review.

SkinSavvy: Skincare Assistant

Technologies: Flutter, Dart, FastAPI, SQLAlchemy, Docker

- Flutter app that reads ingredient lists against the skin profile, catches routine clashes and reminds when opened products expire; a community product database that grows through a team review queue.
- FastAPI server with argon2, rotating refresh tokens and session-family revocation on suspected theft; ~99% test coverage. Live demo: melisayalcinkaya.github.io/SkinSavvy

QR Menu

Technologies: Django REST Framework, React.js, Docker, PySide6

- Digital menu and owner panel for restaurants, opened from table-specific QR codes, with waiter calls and bill requests, 14 allergen filters, Turkish/English content and three themes.
- API secured with JWT, ownership-based authorisation and rate limits; 90+ tests and CI. PySide6 desktop call screen released as a Windows .exe. Live demo: qr-menus.onrender.com

QR Code Attendance System

Technologies: Django, Python, SQLite, Docker

- Cheat-resistant attendance with QR codes that refresh every 5 seconds and expire within 15, an optional location check and one check-in per device; live list and Excel export. Live demo: qr-yoklama-e9u9.onrender.com

ATASENG Club Website

Technologies: React.js, Figma

- Turned a Figma interface design for a student club into an interactive React.js site.

CERTIFICATES

AI & Data Science: Machine Learning Applications with Python, Introduction to Deep Learning, Introduction to Artificial Intelligence, Machine Learning — BTK Akademi

Systems & Game Development: Introduction to Operating Systems, Introduction to Game Development with Unity — BTK Akademi

Web & Software Development: Web Development with HTML5, CSS Fundamentals — BTK Akademi

PUBLICATIONS

Fake News Detection with the Tf-Idf Vector Model and Artificial Intelligence Techniques

Tf-Idf Vektör Modeli ve Yapay Zekâ Teknikleriyle Sahte Haber Tespiti

Karcıoğlu, A. A., Kara, Ş., & Yalçınkaya, M. — Conference paper, ICADA — 14/03/2024

- To fight misinformation, built a machine-learning classifier using text mining and preprocessing techniques on the ISOT dataset.
- Of four algorithms tested with 5-fold cross-validation and a 70/30 split, the Decision Tree model reached 99.24% accuracy.

REFERENCES

Available on request.